

Chapter 1: Elementary function

National High School of Mathematics

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1. Arcsin Function

1.1 Definition

The sine function, $\sin(x)$, is defined for all $x \in \mathbb{R}$, but when restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, it is **continuous, increasing, and bijective** onto $[-1, 1]$.

Thus, we define its **inverse function**, called the **arcsine function** or **arcsin**, as:

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \text{where} \quad x \mapsto \arcsin(x)$$

Since $\sin(x)$ is increasing and differentiable on $(-\frac{\pi}{2}, \frac{\pi}{2})$, its derivative is $\sin'(x) = \cos(x) > 0$. Thus, the inverse function $\arcsin(x)$ is also **continuous and differentiable** in $(-1, 1)$.

1. $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ is **continuous** on its domain and **differentiable** on $(-1, 1)$.
2. $\arcsin(\sin x) = x$ for all $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.
3. $\sin(\arcsin x) = x$ for all $x \in [-1, 1]$.
4. $\arcsin(-x) = -\arcsin(x)$ for all $x \in [-1, 1]$.

• Proof of Property 4

For any $x \in [-1, 1]$, we also have $-x \in [-1, 1]$. Let $\theta = \arcsin(x)$. By definition, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\sin(\theta) = x$. It follows that:

$$-x = -\sin(\theta) = \sin(-\theta)$$

Since $-\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, we can apply the arcsin function to both sides:

$$\arcsin(-x) = \arcsin(\sin(-\theta)) = -\theta$$

Substituting $\theta = \arcsin(x)$ gives the desired result: $\arcsin(-x) = -\arcsin(x)$.

The function $\arcsin(\sin x)$ is **well-defined** for all $x \in \mathbb{R}$ because arcsin is defined on $[-1, 1]$, and $\sin(x)$ always returns a value in this range. However, note that the identity $\arcsin(\sin x) = x$ is only valid for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. For example:

$$\arcsin(\sin 2\pi) = \arcsin(0) = 0 \neq 2\pi$$

1.2 Table of Common Values of Arcsin

Table 1: Common Values of the Arcsin Function

x	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\arcsin(x)$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$

1.3 Differentiability of Arcsin

Since $\sin(\arcsin x) = x$ for all $x \in [-1, 1]$, differentiating both sides gives:

$$\cos(\arcsin x) \cdot (\arcsin x)' = 1$$

Solving for $(\arcsin x)'$:

$$(\arcsin x)' = \frac{1}{\cos(\arcsin x)}$$

Using the identity $\cos^2(\theta) + \sin^2(\theta) = 1$ and noting that $\cos(\theta) \geq 0$ for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, we get:

$$\cos(\arcsin x) = \sqrt{1 - \sin^2(\arcsin x)} = \sqrt{1 - x^2}$$

Thus, the derivative is:

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}, \quad x \in (-1, 1)$$

2. Arccosine Function

2.1 Definition

The function $\cos(x)$ is defined for all $x \in \mathbb{R}$, but when restricted to $[0, \pi]$, it is **continuous, decreasing, and bijective** onto $[-1, 1]$. Its inverse function is called the **arccosine function** or **arccos**:

$$\arccos : [-1, 1] \rightarrow [0, \pi] \quad \text{where} \quad x \mapsto \arccos(x)$$

Since $\cos(x)$ is decreasing and differentiable on $(0, \pi)$, its derivative is $\cos'(x) = -\sin(x) < 0$. Thus, the inverse function $\arccos(x)$ is also **continuous and differentiable** in $(-1, 1)$.

1. $\arccos : [-1, 1] \rightarrow [0, \pi]$ is **continuous and decreasing** on $[-1, 1]$.

2. $\arccos(\cos x) = x$ for all $x \in [0, \pi]$.

3. $\cos(\arccos x) = x$ for all $x \in [-1, 1]$.

4. $\arccos(-x) = \pi - \arccos(x)$ for all $x \in [-1, 1]$.

5. $\arccos(x) + \arcsin(x) = \frac{\pi}{2}$ for all $x \in [-1, 1]$.

6. $\arccos(x)$ is **continuously differentiable** on $(-1, 1)$.

• Proof of Properties 4 and 5

Proof of 4: For any $x \in [-1, 1]$, let $\theta = \arccos(x)$. By definition, $\theta \in [0, \pi]$ and $\cos(\theta) = x$. We know that $\cos(\pi - \theta) = -\cos(\theta) = -x$. Since $\theta \in [0, \pi]$, then $\pi - \theta \in [0, \pi]$. Applying the arccos function:

$$\arccos(-x) = \arccos(\cos(\pi - \theta)) = \pi - \theta = \pi - \arccos(x)$$

Proof of 5: Let $f(x) = \arcsin(x) + \arccos(x)$ for $x \in [-1, 1]$. The function is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. Its derivative is:

$$f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0 \quad \text{for all } x \in (-1, 1)$$

Since the derivative is zero, $f(x)$ is constant on $[-1, 1]$. We evaluate at $x = 0$:

$$f(0) = \arcsin(0) + \arccos(0) = 0 + \frac{\pi}{2} = \frac{\pi}{2}$$

Therefore, $\arcsin(x) + \arccos(x) = \frac{\pi}{2}$ for all $x \in [-1, 1]$.

Since $\cos(x) \in [-1, 1]$ for all $x \in \mathbb{R}$, the expression $\arccos(\cos x)$ is well defined for all real x . However, the identity $\arccos(\cos x) = x$ is true only for $x \in [0, \pi]$.

2.2 Table of Common Values of Arccos

2.3 Differentiability of Arccos

Since $\cos(\arccos x) = x$ for all $x \in [-1, 1]$, differentiating both sides gives:

$$-\sin(\arccos x) \cdot (\arccos x)' = 1$$

Table 2: Common Values of the Arccos Function

x	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\arccos(x)$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$

Solving for $(\arccos x)'$:

$$(\arccos x)' = -\frac{1}{\sin(\arccos x)}$$

Using $\sin(\theta) = \sqrt{1 - \cos^2(\theta)}$ for $\theta \in [0, \pi]$:

$$\sin(\arccos x) = \sqrt{1 - \cos^2(\arccos x)} = \sqrt{1 - x^2}$$

Thus:

$$(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}, \quad x \in (-1, 1)$$

3. Arctangent Function

3.1 Definition

The function $\tan(x)$ is defined for all $x \in \mathbb{R}$ except at $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$. When restricted to the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$, it is continuous, strictly increasing, and bijective onto $\mathbb{R}$. It admits an inverse function, called the **arctangent function** or **arctan**, defined as:

$$\arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \quad \text{where} \quad x \mapsto \arctan(x)$$

Since $\tan(x)$ is smooth and has a nonzero derivative on $(-\frac{\pi}{2}, \frac{\pi}{2})$, its inverse function is also **continuously differentiable**.

Session: 5/2/2025

3.2 Table of Common Values of Arctan

Table 3: Common Values of the Arctan Function

x	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\rightarrow \infty$
$\arctan(x)$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\rightarrow \frac{\pi}{2}$

-
1. $\arctan : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$.
 2. $\arctan$ is increasing.
 3. $\arctan(\tan x) = x$ for all $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$.
 4. $\tan(\arctan x) = x$ for all $x \in \mathbb{R}$.
 5. $\arctan(-x) = -\arctan(x)$ for all $x \in \mathbb{R}$.

3.3 Differentiability of Arctan

The derivative of the arctangent function is given by:

$$(\arctan x)' = \frac{1}{1+x^2}$$

- **Proof**

Since $\tan(x)$ is differentiable, we compute its derivative: $\tan'(x) = 1 + \tan^2(x)$. We start with the identity $\tan(\arctan(x)) = x$. Differentiating both sides with respect to x using the chain rule gives:

$$(\tan(\arctan(x)))' = (x)' \implies \tan'(\arctan(x)) \cdot (\arctan x)' = 1$$

Substituting the formula for the derivative of tangent:

$$(1 + \tan^2(\arctan x)) \cdot (\arctan x)' = 1$$

Since $\tan(\arctan(x)) = x$, we have $\tan^2(\arctan(x)) = x^2$. Substituting this in:

$$(1 + x^2) \cdot (\arctan x)' = 1$$

Solving for the derivative of arctan, we get:

$$(\arctan x)' = \frac{1}{1+x^2}$$

4. Logarithm Function

4.1 Definition

We define the **natural logarithm function** as the unique function $\ln(x)$ that satisfies:

- It is defined for $x > 0$.
- It verifies: $\ln'(x) = \frac{1}{x}$ and $\ln(1) = 0$.

The existence of $\ln(x)$ follows from the Fundamental Theorem of Calculus (as $\ln(x) = \int_1^x \frac{1}{t} dt$), and its uniqueness is established as follows:

• Proof of Uniqueness

Suppose there exist two functions $f(x)$ and $g(x)$ satisfying $f'(x) = g'(x) = \frac{1}{x}$ and $f(1) = g(1) = 0$. Define $h(x) = f(x) - g(x)$. Then:

$$h'(x) = f'(x) - g'(x) = \frac{1}{x} - \frac{1}{x} = 0$$

Since $h'(x) = 0$ on $(0, +\infty)$, $h(x)$ must be a constant. We know:

$$h(1) = f(1) - g(1) = 0 - 0 = 0$$

Thus, $h(x) = 0$ for all $x > 0$, implying $f(x) = g(x)$. Therefore, the logarithm function is unique.

1. **Monotonicity:** $\ln(x)$ is a strictly increasing function on $(0, +\infty)$.
2. **Multiplicative Property:** $\ln(xy) = \ln(x) + \ln(y)$, $\forall x, y > 0$.
3. **Power Rule (Integer Case):** $\ln(x^n) = n \ln(x)$, $\forall n \in \mathbb{N}$, $x > 0$.
4. **Power Rule (Rational Case):** $\ln(x^r) = r \ln(x)$, $\forall r \in \mathbb{Q}$, $x > 0$.
5. **Limits:** $\lim_{x \rightarrow \infty} \ln(x) = +\infty$ and $\lim_{x \rightarrow 0^+} \ln(x) = -\infty$.

• Proof of Properties

1. **Monotonicity:** Since $\ln'(x) = \frac{1}{x} > 0$ for $x > 0$, $\ln(x)$ is strictly increasing.
2. **Multiplicative Property:** Fix $y > 0$ and define $h(x) = \ln(xy) - \ln(x) - \ln(y)$ for $x > 0$. Differentiating with respect to x : $h'(x) = \frac{y}{xy} - \frac{1}{x} - 0 = \frac{1}{x} - \frac{1}{x} = 0$. So, $h(x)$ is constant on $(0, +\infty)$. Evaluating at $x = 1$: $h(1) = \ln(y) - \ln(1) - \ln(y) = 0$. Thus, $h(x) = 0$ for all $x > 0$, proving $\ln(xy) = \ln(x) + \ln(y)$.

3. **Power Rule for Natural Numbers:** Proof by induction on n . Base case ($n = 1$): $\ln(x^1) = 1 \cdot \ln(x)$, which is true. Inductive step: Assume $\ln(x^k) = k \ln(x)$ for some $k \in \mathbb{N}$. Consider $\ln(x^{k+1})$:

$$\ln(x^{k+1}) = \ln(x^k \cdot x) = \ln(x^k) + \ln(x) = k \ln(x) + \ln(x) = (k+1) \ln(x)$$

The property holds for all $n \in \mathbb{N}$.

4. **Power Rule for Rational Numbers:** Let $r = \frac{p}{q}$ with $p \in \mathbb{Z}, q \in \mathbb{N}^*$. Let $y = x^r = x^{p/q}$. Then $y^q = x^p$. Taking the natural log of both sides:

$$\ln(y^q) = \ln(x^p) \implies q \ln(y) = p \ln(x)$$

Solving for $\ln(y)$:

$$\ln(y) = \frac{p}{q} \ln(x) \implies \ln(x^r) = r \ln(x)$$

5. Exponential Function

5.1 Definition

Since $\ln(x) : (0, \infty) \rightarrow \mathbb{R}$ is a strictly increasing, continuous, and bijective function, it has an **inverse function**, which we call the **exponential function**, denoted $\exp(x)$, with domain $\mathbb{R}$ and range $(0, \infty)$.

$$\exp : \mathbb{R} \rightarrow (0, \infty)$$

1. $\exp(x + y) = \exp(x) \cdot \exp(y)$ for all $x, y \in \mathbb{R}$.
2. For all $x \in \mathbb{R}, r \in \mathbb{Q}$, $\exp(rx) = (\exp(x))^r$.
3. **Limits:** $\lim_{x \rightarrow \infty} \exp(x) = +\infty$ and $\lim_{x \rightarrow -\infty} \exp(x) = 0$.

• Proof of Properties

1. Let $A = \exp(x)$ and $B = \exp(y)$. This means $\ln(A) = x$ and $\ln(B) = y$. Using the logarithm property: $\ln(A \cdot B) = \ln(A) + \ln(B) = x + y$. Now, take the exponential of both sides:

$$\exp(\ln(A \cdot B)) = \exp(x + y) \implies A \cdot B = \exp(x + y)$$

Substituting back A and B : $\exp(x) \cdot \exp(y) = \exp(x + y)$.

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2. Let $A = \exp(x)$, so $\ln(A) = x$. Using the logarithm power rule:

$$\ln(A^r) = r \ln(A) = rx$$

Now, take the exponential of both sides:

$$\exp(\ln(A^r)) = \exp(rx) \implies A^r = \exp(rx)$$

Substituting back A : $(\exp(x))^r = \exp(rx)$.

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6. Logarithm and Exponential of General Bases

Let a be a positive real number with $a \neq 1$. We define the **logarithm of base a** as:

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}, \quad \text{for } x > 0$$

The **exponential of base a** , written a^x , is the inverse function of $\log_a(x)$.

6.1 Propositions

1. For all $a > 0, a \neq 1$: $(\log_a x)' = \frac{1}{x \ln(a)}$.
2. For $x > 0$:
 - $\log_a(x)$ is **increasing** if $a > 1$.
 - $\log_a(x)$ is **decreasing** if $0 < a < 1$.
3. For all $x, y > 0$: $\log_a(xy) = \log_a(x) + \log_a(y)$.
4. For all $x > 0$ and for all $c \in \mathbb{R}$: $\log_a(x^c) = c \log_a(x)$.

6.2 Real Power Functions

Let $a > 0$ and $x \in \mathbb{R}$. We can find an explicit formula for the exponential base a , a^x : Let $y = a^x$. Then by definition of the inverse, $x = \log_a(y)$.

$$x = \frac{\ln(y)}{\ln(a)} \implies x \ln(a) = \ln(y)$$

Taking the natural exponential of both sides:

$$\exp(x \ln a) = \exp(\ln y) = y$$

This gives the fundamental definition for real powers:

For all $a > 0$ and for all $r \in \mathbb{R}$, we define:

$$a^r = \exp(r \ln a)$$

For $a, b > 0$ and $x, y \in \mathbb{R}$:

- $(ab)^x = a^x b^x$
- $a^{x+y} = a^x a^y$
- $(a^x)^y = a^{xy}$
- $a^0 = 1$ and $a^1 = a$

7. Hyperbolic Functions

We define the hyperbolic sine, hyperbolic cosine, and hyperbolic tangent functions as follows:

1. **Hyperbolic Sine:** $\sinh(x) = \frac{e^x - e^{-x}}{2}$. This is an odd function with Domain: $\mathbb{R}$ and Range: $\mathbb{R}$.
2. **Hyperbolic Cosine:** $\cosh(x) = \frac{e^x + e^{-x}}{2}$. This is an even function with Domain: $\mathbb{R}$ and Range: $[1, +\infty)$.
3. **Hyperbolic Tangent:** $\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. This is an odd function with Domain: $\mathbb{R}$ and Range: $(-1, 1)$.

7.1 Inverse Hyperbolic Functions

The function $\cosh(x)$ restricted to $[0, +\infty)$ is bijective onto $[1, \infty)$, so it admits an inverse, denoted $\operatorname{arccosh}(x)$ or $\cosh^{-1}(x)$.

The function $\sinh(x)$ is bijective from $\mathbb{R}$ to $\mathbb{R}$, so it admits an inverse, denoted $\operatorname{arsinh}(x)$ or $\sinh^{-1}(x)$.

The function $\tanh(x)$ is bijective from $\mathbb{R}$ to $(-1, 1)$, so it admits an inverse, denoted $\operatorname{artanh}(x)$ or $\tanh^{-1}(x)$.

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7.2 Hyperbolic Identities

- $\cosh^2(a) - \sinh^2(a) = 1$
- $\cosh(a + b) = \cosh(a) \cosh(b) + \sinh(a) \sinh(b)$
- $\sinh(a + b) = \sinh(a) \cosh(b) + \cosh(a) \sinh(b)$
- $\cosh(2a) = \cosh^2(a) + \sinh^2(a) = 2 \cosh^2(a) - 1 = 1 + 2 \sinh^2(a)$
- $\sinh(2a) = 2 \sinh(a) \cosh(a)$

7.3 Inverse Function Formulas

1. $\operatorname{arccosh}(\alpha) = \ln \left(\alpha + \sqrt{\alpha^2 - 1} \right), \quad \alpha \geq 1$
2. $\operatorname{arsinh}(\alpha) = \ln \left(\alpha + \sqrt{\alpha^2 + 1} \right), \quad \alpha \in \mathbb{R}$
3. $\operatorname{artanh}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad x \in (-1, 1)$

- **Derivation of arccosh formula**

Let $y = \operatorname{arccosh}(\alpha)$, where $\alpha \geq 1$ and $y \geq 0$. By definition, $\cosh(y) = \alpha$.

$$\frac{e^y + e^{-y}}{2} = \alpha$$

Multiplying by $2e^y$ gives:

$$e^{2y} + 1 = 2\alpha e^y \implies e^{2y} - 2\alpha e^y + 1 = 0$$

This is a quadratic equation in $u = e^y$. Using the quadratic formula for u :

$$u = \frac{2\alpha \pm \sqrt{4\alpha^2 - 4}}{2} = \alpha \pm \sqrt{\alpha^2 - 1}$$

So, $e^y = \alpha \pm \sqrt{\alpha^2 - 1}$. Since $y \geq 0$, we must have $e^y \geq 1$. For $\alpha > 1$, the term $\alpha - \sqrt{\alpha^2 - 1} <$

1. Therefore, we must choose the positive root:

$$e^y = \alpha + \sqrt{\alpha^2 - 1}$$

Taking the natural logarithm of both sides gives the result:

$$y = \operatorname{arccosh}(\alpha) = \ln \left(\alpha + \sqrt{\alpha^2 - 1} \right)$$

7.4 Derivatives of Inverse Hyperbolic Functions

1. $\frac{d}{d\alpha} \operatorname{arccosh}(\alpha) = \frac{1}{\sqrt{\alpha^2-1}}, \quad \alpha > 1$
2. $\frac{d}{d\alpha} \operatorname{arcsinh}(\alpha) = \frac{1}{\sqrt{\alpha^2+1}}, \quad \alpha \in \mathbb{R}$
3. $\frac{d}{dx} \operatorname{arctanh}(x) = \frac{1}{1-x^2}, \quad x \in (-1, 1)$

- **Derivation of arccosh derivative**

We differentiate the explicit formula $\operatorname{arccosh}(\alpha) = \ln(\alpha + \sqrt{\alpha^2 - 1})$. Using the chain rule:

$$\begin{aligned} \frac{d}{d\alpha} \operatorname{arccosh}(\alpha) &= \frac{1}{\alpha + \sqrt{\alpha^2 - 1}} \cdot \frac{d}{d\alpha} (\alpha + \sqrt{\alpha^2 - 1}) \\ &= \frac{1}{\alpha + \sqrt{\alpha^2 - 1}} \cdot \left(1 + \frac{1}{2\sqrt{\alpha^2 - 1}} \cdot 2\alpha \right) \\ &= \frac{1}{\alpha + \sqrt{\alpha^2 - 1}} \cdot \left(1 + \frac{\alpha}{\sqrt{\alpha^2 - 1}} \right) \\ &= \frac{1}{\alpha + \sqrt{\alpha^2 - 1}} \cdot \left(\frac{\sqrt{\alpha^2 - 1} + \alpha}{\sqrt{\alpha^2 - 1}} \right) \\ &= \frac{1}{\sqrt{\alpha^2 - 1}} \end{aligned}$$

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